



Author: HAF

Note from author

These notes are mostly based on the syllabus and past papers

Fair warning: they might look short (2–3 pages), but I've condensed everything down to keep the important stuff while removing the extra yap. That said, I wouldn't recommend using these as quick revision notes, but think of them as detailed notes covering almost everything

Spotted a mistake? Suggestions?

I'd love to hear your feedback. Contact us at: learnmates.share@gmail.com

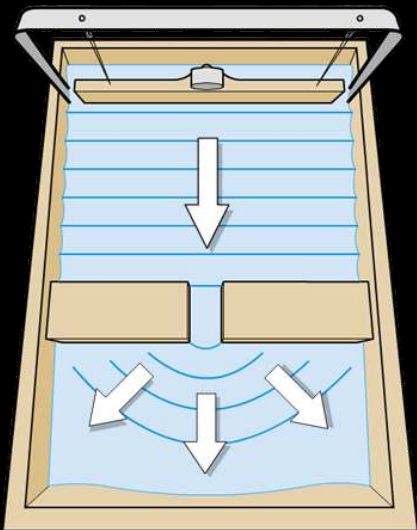
For more resources visit learnmates.org

Diffraction

Diffraction is the process by which a wave incident at slit or obstacle spreading into geometrical shadow (this would also be used to describe the diffraction in a grating or double slit)



Demonstrating diffraction



Using a ripple tank:

• Vibrator attached to a dipper bar acts as the source, generating parallel plane waves on the water's surface which then encounter a barrier with a narrow slit. A stroboscope positioned above the tank serves as the detector, and when its flash rate is synchronized with the vibrator, it effectively "freezes" the wave motion for clear observation.

Observation:

• diffraction: the plane waves do not continue in a straight line but instead bend and spread out circularly from the slit into the region behind the barrier, with the effect being most pronounced when the width of the slit is approximately equal to the wavelength.

Gap size and its affect on diffraction

If gap size $>$ wavelength (λ)

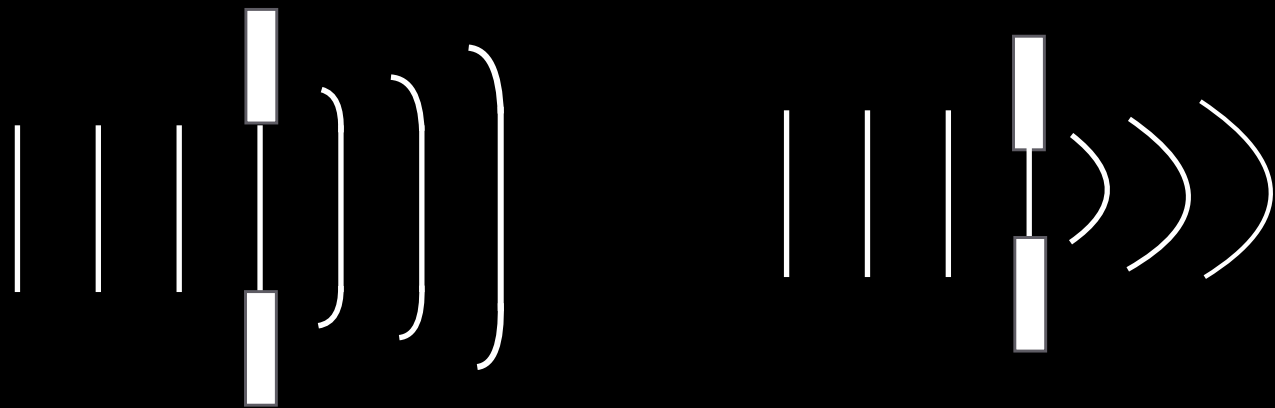
Diffraction is minimal.

Most of the wave passes straight through and the wave only spreads out slightly at the edges of the gap.

If gap size $\approx$ wavelength (λ)

Diffraction is noticeable and efficient

The wave spreads out significantly into the region behind the obstacle/gap. This is the optimal condition for studying diffraction effects.



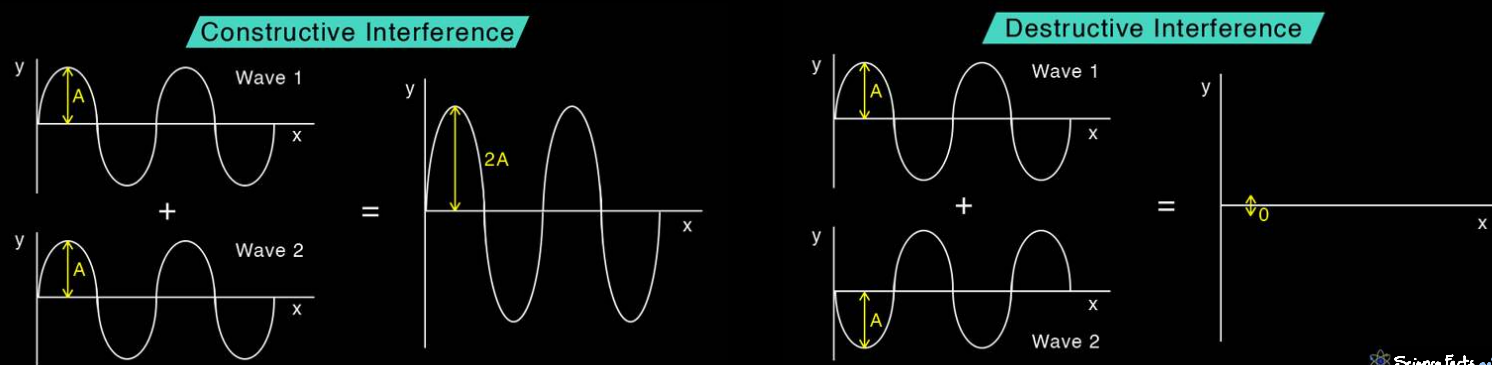
Additional thing to note:

As the gap size decreases, even though the diffraction effect (spreading) increases but the amount of waves that pass through decreases (most of them get reflected or absorbed)

Which gets us to our net point, if the gaps size $<$ wavelength (λ) $\rightarrow$ the amount of diffraction remains almost the same but the amount of passing waves decreases (hence less intense)

Superposition

Principle of superposition: When two or more waves meet/overlap at a point the resultant displacement is sum of the individual displacements.



Interference: Phenomenon that occurs when two or more waves overlap and superpose where the resultant displacement is the sum of the individual displacements.

For two waves to interfere:

They must overlap/ meet

They must be the same type (e.g. can't interfere with light waves)

They must have the same plain of polarizations

Phase difference

We have already talked about phase difference between two waves.

Causes of phase difference between two waves:

1 - Initial Phase Difference (From the Sources)

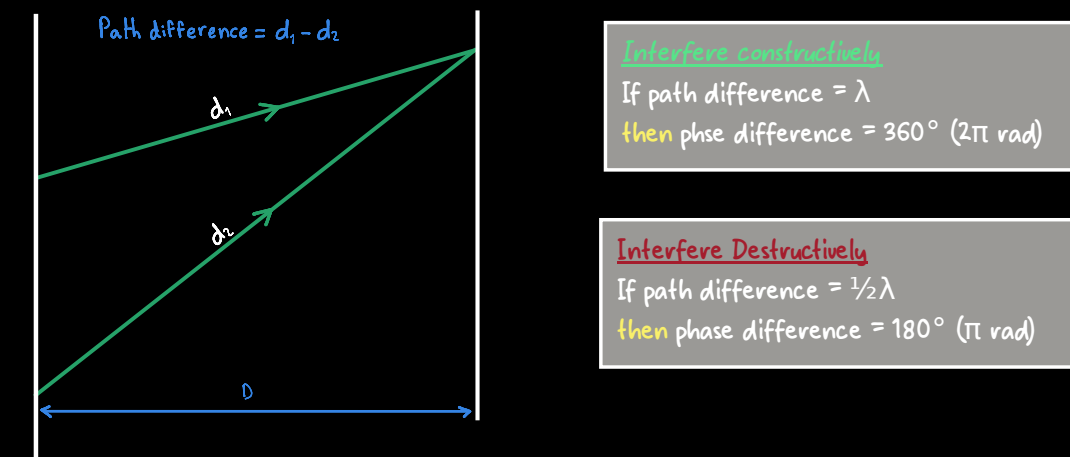
$\rightarrow$ The two sources themselves might not start in phase with each other

2 - Path Difference (Most Important for Double Slit later on)

Path difference: is the extra distance one wave travels compared to the other.

$\rightarrow$ Waves travel through space at a fixed speed $\rightarrow$ If one wave travels further $\rightarrow$ it takes more time to arrive $\rightarrow$ it is out of phase with the other wave

So even though the wave might be in phase initially they would become out of phase

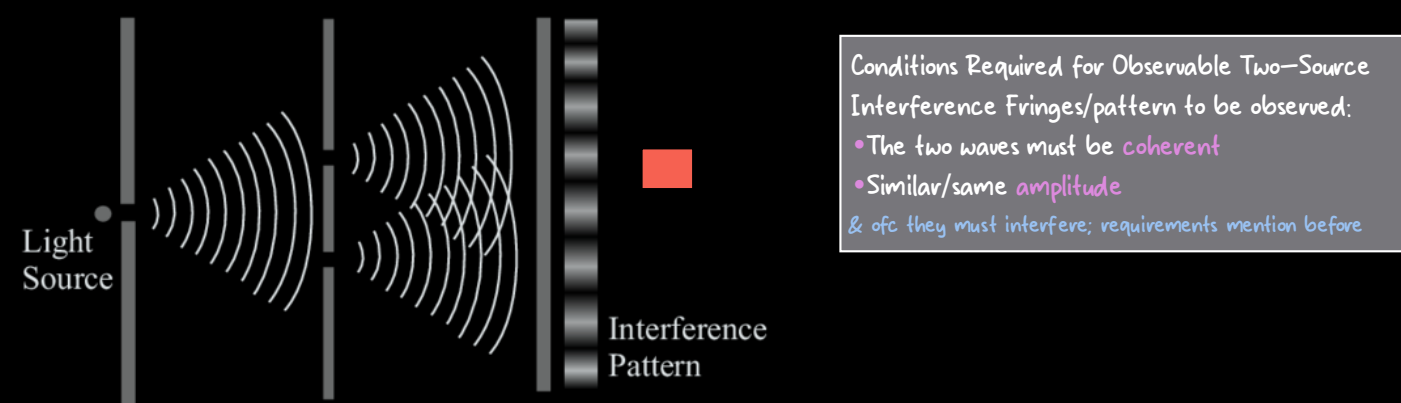


Two-Source Interference

Two-source interference (double slit) creates two coherent wave sources and allowing their waves to overlap, we can observe the interference pattern directly. This works for all types of waves: water waves, sound waves, light waves, and microwaves.

Process:

- A single source is used to ensure coherent waves are produced
- The wave is diffracted by the double slit into two waves, where the wave at each slit spread $\rightarrow$ so they overlap
- At each point where waves meet, superposition occurs (they add together)
- The path difference from each source to that point is compared:
If path difference = $n\lambda \rightarrow$ waves arrive in phase $\rightarrow$ constructive interference $\rightarrow$ maximum (bright/loud)
If path difference = $(n + \frac{1}{2})\lambda \rightarrow$ waves arrive out of phase $\rightarrow$ destructive interference $\rightarrow$ minimum (dark/quiet)
- This creates an interference pattern of alternating maxima and minima
- The pattern remains stable because the sources are coherent



Explain Interference Observations

Step 1: Find the Path Difference: Path difference (PD) = |distance from source 1| - |distance from source 2|

Calculate or state the path difference between the two waves arriving at that point.

then convert it in terms of λ : using PD / wavelength (e.g. PD = 3 and $\lambda = 6 \rightarrow 3/6 = \frac{1}{2}$, so PD = $\frac{1}{2}\lambda$)

Step 2: Convert to Phase Difference

Use the relationship between path difference and phase difference

Multiply ratio (PD / λ) $\times 360^\circ$ (or 2π rad)

(e.g. if PD/ $\lambda = 15 \rightarrow 15 \times 360 = 540^\circ$)

Step 3: State the Observation

Based on phase difference, state whether interference is constructive or destructive and describe what is observed.

0° or $360n^\circ \rightarrow$ Constructive - Observation: Bright fringe (light) / Loud sound / Large amplitude (water) / high intensity

180° or $360n^\circ \rightarrow$ Destructive - Observation: Dark fringe (light) / Quiet or silent (sound) / Flat water (ripple tank) / low intensity (where $n = 0, 1, 2, 3, \dots$)

Note: If the phase difference is not exactly $0^\circ/360n^\circ$ (constructive) or $180^\circ + 360n^\circ$ (destructive), the waves still interfere, but the resulting amplitude/intensity will be between the maximum (constructive) and minimum (destructive) values.

The Double Slit Formula

Formula: $x = \frac{\lambda D}{a}$

λ - Wavelength

a - Slit separation (distance between the centres of the two slits)

x - Fringe spacing (distance between adjacent bright fringes)

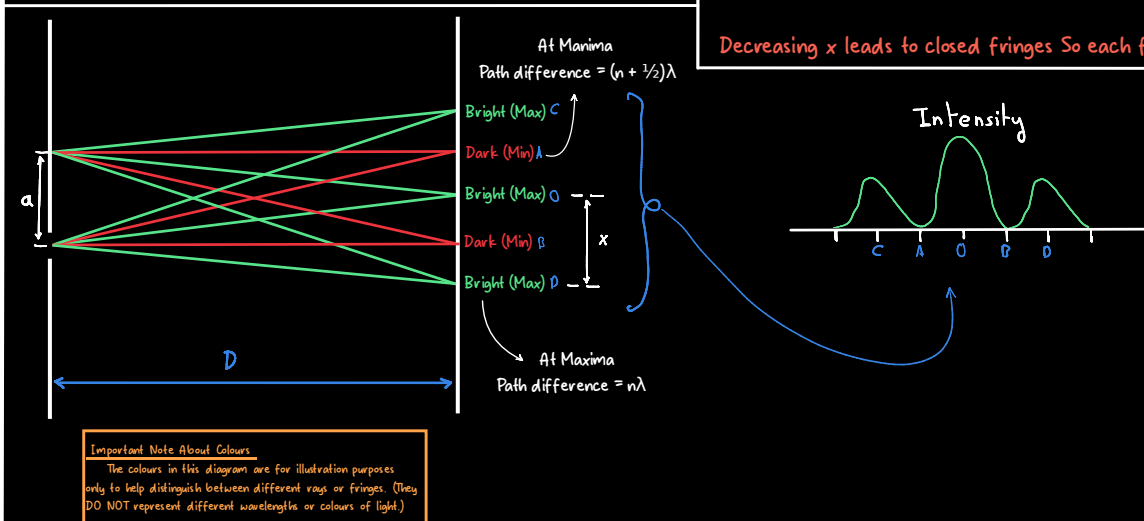
D - Distance from slits to screen

Effects on the Interference Pattern $x = \frac{\lambda D}{a}$

$a \uparrow$ (slits further apart) $\rightarrow x \downarrow$ [Closer together]
 $D \uparrow$ (screen further) $\rightarrow x \uparrow$ [More spread out] & less bright
 $\lambda \uparrow$ (e.g. red) $\rightarrow x \uparrow$ [More spread out]

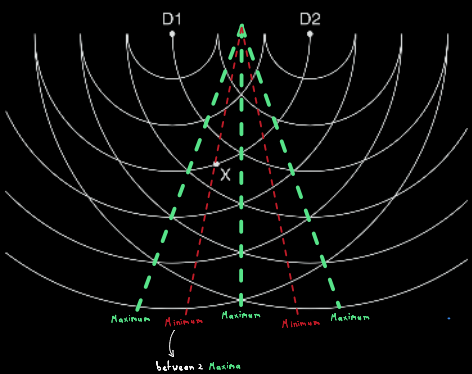
Slit width $\uparrow \rightarrow$ Intensity (Brightness of light fringes) $\uparrow$
 Source intensity $\uparrow \rightarrow$ Intensity (Brightness of light fringes) $\uparrow$

Decreasing x leads to closer fringes so each fringe is dimmer (vice versa)



Experiments demonstrating demonstrate two-source interference

Water Waves (Ripple Tank)



Aspect	Details
Setup	- Ripple tank with water - Two dippers attached to the same vibrating bar - Dippers touch water surface and create circular waves - Stroboscope may be used to 'freeze' the pattern
What happens	Circular waves from each dipper overlap and superpose
Observation	- Constructive interference: Large amplitude ripples (bright regions under lighting) - Destructive interference: Flat water (dark regions)
Why it works	Same vibrating bar ensures dippers produce waves with same frequency and constant phase difference → coherent sources

Sound

Aspect	Details
Setup	- Two loudspeakers connected to the same signal generator - Speakers placed side by side, facing forward - Signal generator set to a single frequency (pure tone)
What happens	Sound waves from each speaker spread out and overlap
Observation	- Walk a microphone (or your ear) across the room in front of the speakers - You hear loud spots (constructive interference) and quiet spots (destructive interference) - Pattern repeats at regular intervals
Why it works	Same signal generator → both speakers produce waves with same frequency and constant phase difference → coherent sources

Microwave

Aspect	Details
Setup	- Microwave transmitter - Two slits in a metal barrier (or two separate horns fed from same source) - Microwave probe detector connected to meter
What happens	Microwaves diffract through slits and overlap
Observation	- Move detector across the space behind the slits - Meter shows maxima (constructive) and minima (destructive) in signal strength
Why it works	Single transmitter feeds both slits → coherent sources

Diffraction Grating

A diffraction grating is a plate with many closely spaced, parallel slits etched into it. Typically hundreds or thousands of slits per millimetre. Typical slit spacing: $d \approx 10^{-6}$ m (for visible light). When light passes through it, each slit acts as a coherent source of waves due to diffraction.

Diffraction grating vs double slit

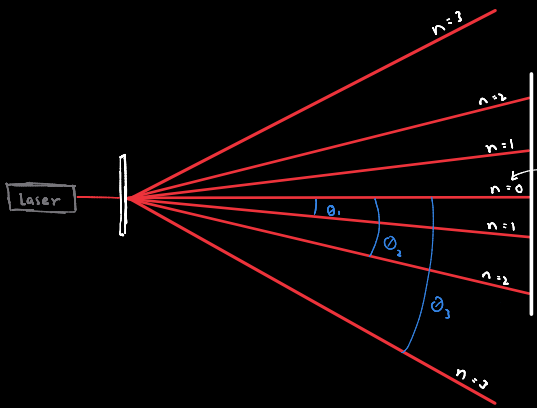
Because there are many slits, the maxima are:

- Much brighter (more slits to pass through + more waves contributing)
- Much sharper/narrower (precise angles)

Unlike a double slit, the interference pattern from a diffraction grating is much more precise:

- Waves interfere constructively only at specific angles where $d \sin \theta = n \lambda$
- At all other angles, waves from the many slits interfere destructively and cancel out completely

This is because with thousands of slits, for any angle slightly away from a maximum, there will always be another slit whose wave is exactly out of phase to cause cancellation.



Order??

Orders are the directions in which waves from all slits interfere constructively where each order corresponds to a path difference of $n \lambda$ between waves from adjacent slits. They are the numbered lines that appear at specific angles on either side of the central maximum.

- $n = 0$ → Central maximum (straight through)
- $n = 1$ → First maximum on either side
- $n = 2$ → Second maximum, further out
- $n = 3$ → Third maximum, even further

Finding the Maximum Order

The max possible angle for an order is $90^\circ \rightarrow \sin(90^\circ) = 1$. Find n using: $n = \frac{\lambda}{d} \rightarrow$ round n down to the nearest integer.

The Diffraction Grating Formula

Formula: $d \sin \theta = n \lambda$

d – Slit separation (distance between adjacent slits)
 θ – Angle from the central maximum ($n=0$) to the n th order maximum
 n – Order of the maximum/diffraction (0, 1, 2, 3...)
 λ – Wavelength of the light

Where the Formula Comes From

Path difference between waves from adjacent slits = $d \sin \theta$
For constructive interference: path difference = $n \lambda$
Therefore: $d \sin \theta = n \lambda$

White Light (Multiple Colours) in Interference

White light contains all visible wavelengths (approximately 400 nm violet to 700 nm red). From $x = \frac{\lambda D}{d}$ (double slit) or $d \sin \theta = n \lambda$ (grating) we can deduce that for a greater wavelength λ there will be more spread (larger x or θ) and vice versa. [Each color (λ) would have its own path difference that will lead to constructive interference.]

Observations

Central ($n=0$): White

Reason: All wavelengths have zero path difference → all construct together

First order: Spectrum with blue/violet nearest centre, red farthest out

Reason: $x \propto \lambda \rightarrow$ red spreads more than blue

Higher orders:

Both: intensity decreases at higher orders

Double slit: Blurred and overlapping → becomes broader and more indistinct

Explanation: Each color's fringe (maximum) is wide → Colours smear together → pattern loses definition

Diffraction grating: Spectra may overlap but still clear and sharp colors

Explanation: Each color's fringe (maximum) is narrow → Constructive interference only at exact angles

→ colours stay distinct, and lines (each wavelength) become further apart

The double slit and diffraction grating formulas assume incident light rays are perpendicular (normal) to the plane of the slits.

This ensures no initial path difference, and the only path difference comes after diffraction.

Stationary (Standing) Waves

A stationary wave (also called a standing wave) is a wave pattern that remains in a fixed position. It does not appear to travel like a progressive wave.

Formation:

- Two waves of the same frequency, speed, amplitude and type travel in opposite directions and overlap
- The resultant displacement is the sum of the displacements of each wave
- This produces a pattern of nodes and antinodes
- Usually these two waves are the incident wave and its reflection

Stationary Waves vs Progressive Waves

Stationary Wave	Progressive Wave
Has nodes and antinodes	Does not have nodes and antinodes
Does not transfer/propagate energy	Transfers/propagates energy
Different points have different amplitudes	All points have the same/constant amplitude

Points on a stationary waves

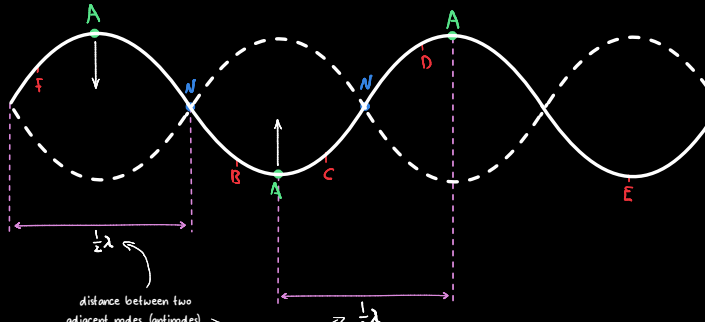
There are two main types of point on a wave:

Node (N): Points that remain stationary with zero displacement (amplitude)

Cause: The two waves meet antiphase (destructively)

Antinodes (A): Point with max displacement (amplitude)

Cause: The two waves meet inphase (constructively)



Phase difference between two points:

– If there is an odd number of node between them → these points are antiphase

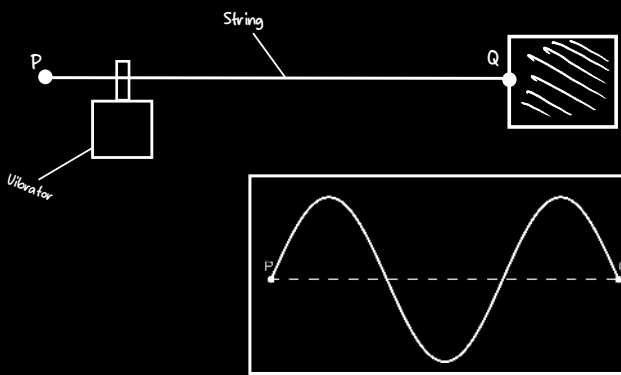
– If there is an even number of node between them → these points are inphase

B is inphase with C and E
B is antiphase with F and D

Experiments Demonstrating Stationary Waves and determining wavelengths

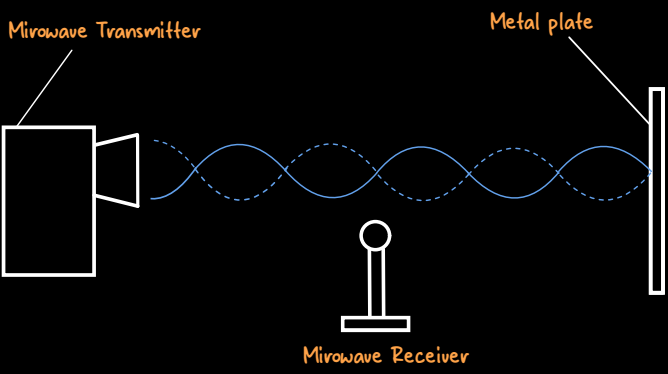
Stretched String

Aspect	Details
Setup	- String fixed at both ends - Vibration generator attached near one end - Frequency varied until resonance occurs
What happens	- Wave travels along string, reflects at fixed ends - Incident and reflected waves travel in opposite directions → superpose → form stationary wave
Observation	- Nodes at both fixed ends (cannot move) - Antinodes form at points of maximum vibration - Clear loops appear at resonance
Finding λ	- Measure total length L between fixed ends - Identify harmonic number n - Use $\lambda = \frac{2L}{n}$



Microwaves

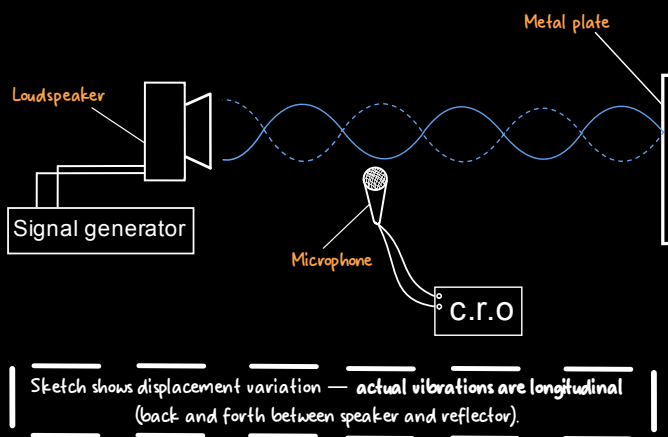
Aspect	Details
Setup	- Microwave transmitter - Metal reflector plate - Microwave probe detector connected to a meter
What happens	- Waves travel to reflector and reflect back - Incident and reflected waves travel in opposite directions → superpose → form stationary wave
Observation	- Antinode at transmitter - Node at reflector (due to 180° phase change) - Detector shows alternating maxima (antinodes) and minima (nodes)
Finding nodes and antinodes	- Move detector slowly from transmitter to reflector - Maxima on meter = antinodes - Minima on meter = nodes
Finding λ	- Measure distance between two adjacent minima (nodes) - This distance = $\lambda/2 \rightarrow \lambda = 2 \times$ node distance (Repeat for several pairs of adjacent minima and average the results to reduce error)



Sound waves

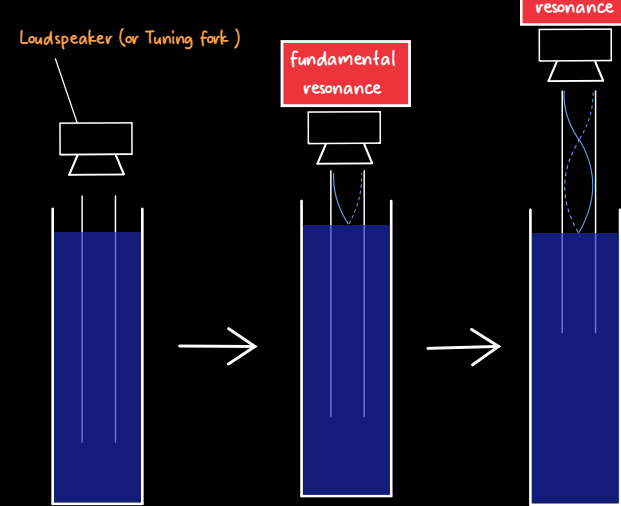
Aspect	Details
Setup	- Loudspeaker connected to a signal generator - Rigid reflector (e.g. metal plate or wall) - Microphone connected to a cathode ray oscilloscope (CRO) or sound level meter
What happens	- Sound waves travel to reflector and reflect back - Incident and reflected waves travel in opposite directions → superpose → form stationary wave
Observation	- Antinode near loudspeaker - Node at reflector (due to 180° phase change) - Detector shows alternating maxima (antinodes) and minima (nodes)
Finding nodes and antinodes	- Move microphone slowly from loudspeaker to reflector - Maxima on CRO/meter = antinodes (loudest sound) - Minima on CRO/meter = nodes (quietest sound)
Finding λ	- Measure distance between two adjacent minima (nodes) - This distance = $\lambda/2 \rightarrow \lambda = 2 \times$ node distance (Repeat for several pairs of adjacent minima and average the results to reduce error)

Method 1



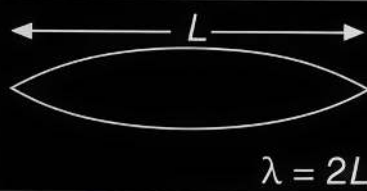
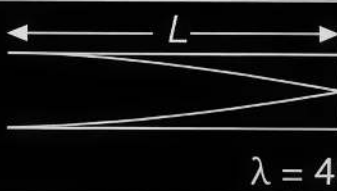
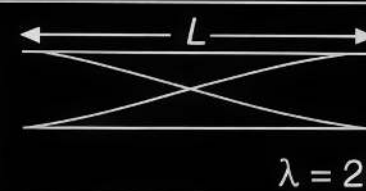
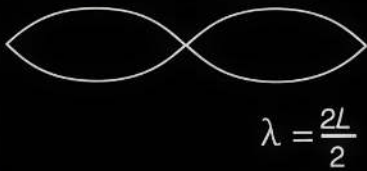
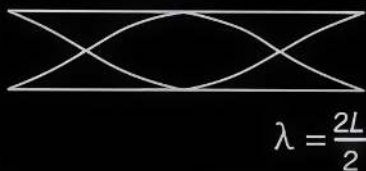
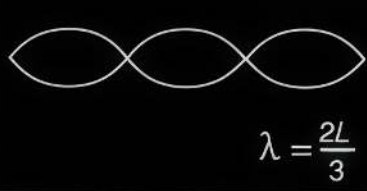
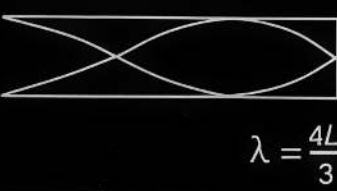
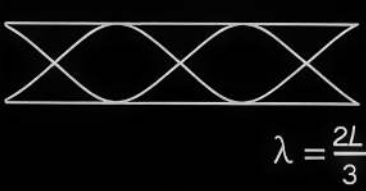
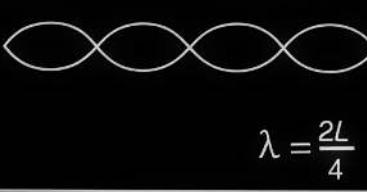
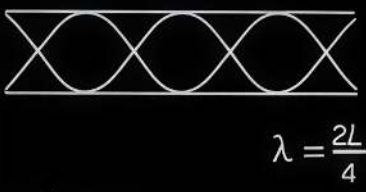
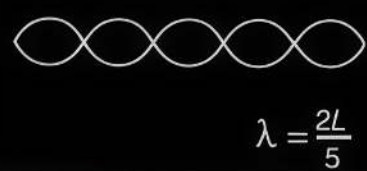
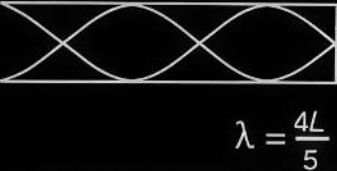
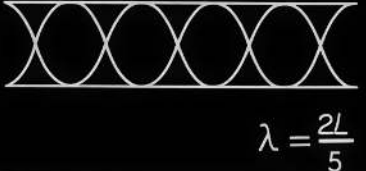
Sketch shows displacement variation — actual vibrations are longitudinal (back and forth between speaker and reflector).

Method 2



Sketch shows displacement variation — actual vibrations are longitudinal (back and forth between speaker and reflector).

Setup	- Vertical tube partially filled with water - Water level in the tube can be adjusted by raising/lowering the tube (or running down water) - Tuning fork of known frequency placed above open end - Ear or microphone used to detect loudness
What happens	- Sound waves travel down tube and reflect at water surface - Incident and reflected waves travel in opposite directions → superpose → form stationary wave - Water surface acts as a node (rigid boundary)
Observation	- Node at water surface - Antinode at open end (air free to vibrate) - Loudest sound (resonance) occurs when stationary wave forms
Finding resonance	- Adjust water level until a loud sound is heard - This indicates an antinode at open end and node at water surface
Finding λ	- Measure length of air column L from water surface to open end - For fundamental ($n=1$): $L = \frac{\lambda}{4} \rightarrow \lambda = 4L$ - For higher resonances: $L = \frac{(2n-1)\lambda}{4} \rightarrow \lambda = \frac{4L}{2n-1}$ - Repeat for different tuning forks or water levels
Finding the Speed of Sound	- For each frequency, adjust water level to find fundamental resonance (loudest sound) - Measure the air column length L at resonance - Draw a line of best fit through the origin - Calculate the gradient $m = \frac{\Delta f}{\Delta (1/L)}$ - Use $m = \frac{v}{4} \rightarrow v = 4m$

Mode	String	Closed Pipe	Open Pipe
1st harmonic or fundamental	 $\lambda = 2L$	 $\lambda = 4L$	 $\lambda = 2L$
2nd harmonic or 1st overtone	 $\lambda = \frac{2L}{2}$	No stationary wave forms — only odd harmonics (1st, 3rd, 5th, 7th...) are possible (node–antinode ends).	 $\lambda = \frac{2L}{2}$
3rd harmonic or 2nd overtone	 $\lambda = \frac{2L}{3}$	 $\lambda = \frac{4L}{3}$	 $\lambda = \frac{2L}{3}$
4th harmonic or 3rd overtone	 $\lambda = \frac{2L}{4}$	No stationary wave forms — only odd harmonics (1st, 3rd, 5th, 7th...) are possible (node–antinode ends).	 $\lambda = \frac{2L}{4}$
5th harmonic or 4th overtone	 $\lambda = \frac{2L}{5}$	 $\lambda = \frac{4L}{5}$	 $\lambda = \frac{2L}{5}$

